

1. (Fall 2024, #7) Given the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2) \cdot 2^n} (x-5)^n$$

- (a) find its radius of convergence,

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{(-1)^{n+1} (x-5)^{n+1}}{(n+3) 2^{n+1}}}{\frac{(-1)^n (x-5)^n}{(n+2) 2^n}} \right| = \left| \frac{n+2}{2(n+3)} (x-5) \right|.$$

$$\xrightarrow{n \rightarrow \infty} \frac{1}{2} |x-5| < 1 \quad \text{converge} \Rightarrow |x-5| < 2$$

$$\Rightarrow \text{radius of convergence } R = 2$$

- (b) find its interval of convergence (that is, the set of points where it is convergent),

Interval of convergence:

$$\begin{aligned} |x-5| < 2 &\Rightarrow -2 < x-5 < 2 \\ &\Rightarrow 3 < x < 7. \end{aligned}$$

$$\left. \begin{aligned} & \\ & \\ & \end{aligned} \right\} \Rightarrow (3, 7).$$

check the end points.

$$\text{when } x=3: \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2) \cdot 2^n} \cdot (-2)^n = \sum_{n=0}^{\infty} \frac{1}{n+2} = \infty \quad \text{divergent}$$

$$\text{when } x=7: \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2) \cdot 2^n} \cdot 2^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+2} < \infty, \text{ by alternating test.}$$

- (c) and find an upper bound for the remainder $|R_5|$ at the only point where the series converges conditionally.

The point is $x=7$.

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n+2} := \sum_{n=0}^{\infty} (-1)^n b_n.$$

By alternating series estimation theorem

$$|R_5| \leq b_6 = \frac{1}{6+2} = \frac{1}{8}.$$

2. (Spring 2024, #8) Consider the function m defined by the power series

$$m(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{2n+1} = 1 - \frac{x^4}{3} + \frac{x^8}{5} - \frac{x^{12}}{7} + \frac{x^{16}}{9} - \dots$$

- (a) Find the radius of convergence of m .

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{(-1)^{n+1} x^{4(n+1)}}{2(n+1)+1}}{\frac{(-1)^n x^{4n}}{2n+1}} \right| = \left| x^4 \cdot \frac{2n+1}{2n+3} \right| \xrightarrow{n \rightarrow \infty} |x|^4 < 1$$

$$\Rightarrow |x| < 1 \Rightarrow \text{radius of convergence } R=1.$$

- (b) Find the interval of convergence of m .

x converges on $(-1, 1)$.
 Check the two end points.
 $x = -1$: $m(-1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} < \infty$ by alternating test.
 $x = 1$: $m(1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} < \infty$, by alternating test.

$\Rightarrow$ Interval of convergence $[-1, 1]$.

- (c) Find $m^{(24)}(0)$.

$m^{(24)}(0)$ = the coefficient of x^{24} , which is the coefficient when $n=6$.

$$\Rightarrow m^{(24)}(0) = \frac{(-1)^6}{2 \cdot 6 + 1} = \frac{1}{13}.$$

- (d) Find the value of

$$\int_0^{1/2} m(x) dx$$

with an error of less than $0.001 = 1/1000$. Leave your answer as a finite sum of fractions.

$$\begin{aligned} \int_0^{1/2} m(x) dx &= \int_0^{1/2} \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cdot \int_0^{1/2} x^{4n} dx \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cdot \frac{1}{4n+1} \left(\frac{1}{2}\right)^{4n+1} = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{1}{(2n+1)(4n+1)2^{4n+1}} \end{aligned}$$

By alternating series estimate theorem, $R_n \leq \frac{1}{(2(n+1)+1)(4(n+1)+1)2^{4(n+1)+1}} \leq \frac{1}{1000}$ choose $n=2$ ✓.

- (e) What is the function $m(x)$? Find an expression for $m(x)$ using familiar functions.

Note that

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^{2n+1} x^{2n+1}}{2n+1}$$

$$\Rightarrow \arctan(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^{2n+1} x^{4n+2}}{2n+1}$$

$$\Rightarrow \frac{\arctan(x^2)}{x^2} = \sum_{n=0}^{\infty} \frac{(-1)^{2n+1} x^{4n}}{2n+1} = m(x).$$

Then the estimate is $\sum_{n=0}^2 (-1)^n \frac{1}{(2n+1)(4n+1)2^{4n+1}}$.

3. (Spring 2024, #10)

- (a) A power series $\sum_{n=0}^{\infty} a_n(x-7)^n$ converges when $x=3$ and diverges when $x=12$. Does the series converge when $x=8$? Circle the best option and explain briefly.

i. It converges absolutely.

ii. It converges though it may only converge conditionally.

iii. It diverges.

iv. There is not enough information to determine this.

Suppose the Radius of convergence is R .

Then in $(7-R, 7+R)$ the series absolutely converges.

In $(-\infty, 7-R)$ $(7+R, +\infty)$, the series diverges.

At $7-R$ and $7+R$, conditionally converge OR diverge.

From the condition. $x=3$ converge $\Rightarrow 3 \geq 7-R$.

$x=12$ diverge $\Rightarrow 12 \geq 7+R$

$\Rightarrow 4 \leq R \leq 5$, $\Rightarrow x=8 \in (7-R, 7+R)$. $\Rightarrow$ converges absolutely.

- (b) Consider the power series $\sum_{n=0}^{\infty} a_n(x-9)^n$. Suppose

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{7^n} = 4.$$

What is the radius of convergence of the power series, R ? Choose one option and explain briefly.

i. $1/7$

ii. $1/4$

iii. 4

iv. 7

v. 9

vi. There is no enough information to determine this.

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}(x-9)^{n+1}}{a_n(x-9)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \cdot |x-9|$$

$$\text{Note that } \lim_{n \rightarrow \infty} \frac{|a_n|}{7^n} = 4, \quad \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{7^{n+1}} = 4.$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{\frac{|a_{n+1}|}{7^{n+1}}}{\frac{|a_n|}{7^n}} \cdot 7 = \frac{4}{4} \cdot 7 = 7$$

$$\Rightarrow L = 7 \cdot |x-9| < 1 \Rightarrow |x-9| < \frac{1}{7} \Rightarrow R = \frac{1}{7}.$$

4. (Fall 2023, #7)

(a) Find the interval of convergence of the series

$$\sum_{n=2}^{\infty} \frac{(-2x)^n}{n^{3/2} \cdot \ln(n+1)}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{(-2x)^{n+1}}{(n+1)^{3/2} \cdot \ln(n+2)}}{\frac{(-2x)^n}{n^{3/2} \cdot \ln(n+1)}} \right| = \left| 2x \cdot \frac{\ln(n+2)}{\ln(n+1)} \cdot \left(\frac{n}{n+1} \right)^{3/2} \right| \xrightarrow{\text{as } n \rightarrow \infty} 2|x|.$$

Then $2|x| < 1 \Rightarrow |x| < \frac{1}{2} \Rightarrow$ On $(-\frac{1}{2}, \frac{1}{2})$ absolutely convergent.

check the end points $x = -\frac{1}{2}$ & $x = \frac{1}{2}$.

When $x = \frac{1}{2} \Rightarrow \sum_{n=2}^{\infty} \frac{(-1)^n}{n^{3/2} \ln(n+1)}$ since $\frac{1}{n^{3/2} \ln(n+1)}$ decreasing to 0. By alternating test, convergent.

When $x = -\frac{1}{2} \Rightarrow \sum_{n=2}^{\infty} \frac{1}{n^{3/2} \ln(n+1)} \leq \sum_{n=2}^{\infty} \frac{1}{n^{3/2} \ln 3}$, since $\frac{3}{2} > 1$, by p-test, convergent.

$\Rightarrow$ Convergent on $[-\frac{1}{2}, \frac{1}{2}]$.

(b) Assume the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ is R , where $a_n \geq 0$. Does the series $\sum_{n=1}^{\infty} a_n \sqrt{n} x^{n-1}$ converge for $0 \leq x < R$? Justify your answer.

Yes.

The radius of convergence of $\sum_{n=1}^{\infty} a_n x^n$

$$L = \frac{|a_{n+1} x^{n+1}|}{|a_n x^n|} = \left| \frac{a_{n+1}}{a_n} \right| \cdot |x|$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| |x| < 1 \Rightarrow |x| < \frac{1}{\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|} = R.$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \overset{a_n \geq 0}{=} \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{1}{R}.$$

Now, look at $\sum_{n=1}^{\infty} a_n \sqrt{n} x^{n-1}$, By ratio test

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1} \sqrt{n+1} x^n}{a_n \sqrt{n} x^{n-1}} \right| = \frac{1}{R} \cdot |x| < 1 \text{ for } 0 \leq x < R.$$

$\Rightarrow$ Then by ratio test, the series converges.

5. (Fall 2023, #8)

(a) Find a power series representation of $f(x) = (1+x)^3 \cdot \ln(2+x)$ and its radius of convergence.

Recall, we have $\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n$ for $|t| < 1$. $\rightarrow$ Radius of convergence.

$$(\ln(2+x))' = \frac{1}{2+x} = \frac{1}{2} \cdot \frac{1}{1+\frac{x}{2}} = \frac{1}{2} \cdot \frac{1}{1-(-\frac{x}{2})}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} \cdot x^n$$

with Radius of convergence: $|-x/2| < 1 \Rightarrow |x| < 2 \Rightarrow R = 2$.

$$\text{Then } \ln(2+x) = \int \frac{1}{2+x} dx = \int \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^n dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} \int x^n dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} \cdot \frac{x^{n+1}}{n+1} + C$$

$$\Rightarrow \ln 2 = C \Rightarrow \ln(2+x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{2^{n+1} \cdot (n+1)} + \ln 2$$

(b) Find $f^{(20)}(-1)$.

$$\Rightarrow f(x) = (1+x)^3 \ln(2+x)$$

$$= (1+x)^3 \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{2^{n+1} (n+1)} + \ln 2 \right)$$

$$= (x^3 + 3x^2 + 3x + 1) \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{2^{n+1} (n+1)} + \ln 2 \right)$$

(b) To find $f^{(20)}(-1)$, we need to expand $\ln(2+x)$

at $x = -1$, That is $\ln(2+x) = \ln(1+x+1)$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} (1+x)^{n+1}$$

$$\text{Then } f(x) = (1+x)^3 \ln(2+x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} (1+x)^{n+4}$$

$$\Rightarrow f^{(20)}(-1) \text{ is the coef. of } (1+x)^{20} \Rightarrow n=16 \Rightarrow f^{(20)}(-1) = \frac{(-1)^{16}}{16+1} = \frac{1}{17}$$